

- 19** An acidified solution contains CaCl_2 , FeCl_2 and MnCl_2 , each of concentration 0.10 mol dm^{-3} . Carbon dioxide is blown through the solution until it is saturated with carbon dioxide at 25°C . The concentration of $\text{CO}_3^{2-}(\text{aq})$ in the saturated solution reaches $1 \times 10^{-9} \text{ mol dm}^{-3}$.

The value of the solubility product, K_{sp} , of each of the carbonates at 25°C is given below.

$$\text{CaCO}_3 \quad 3.8 \times 10^{-9} \text{ mol}^2 \text{ dm}^{-6}$$

$$\text{FeCO}_3 \quad 3.2 \times 10^{-11} \text{ mol}^2 \text{ dm}^{-6}$$

$$\text{MnCO}_3 \quad 2.5 \times 10^{-13} \text{ mol}^2 \text{ dm}^{-6}$$

Which statement describes what happens in the solution?

- A** Only CaCO_3 and FeCO_3 are precipitated.
- B** Only CaCO_3 is precipitated.
- C** Only MnCO_3 and FeCO_3 are precipitated.
- D** Only MnCO_3 is precipitated.